Virtual to Real

CANVAS: Commonsense-Aware Navigation System for Intuitive Human-Robot Interaction

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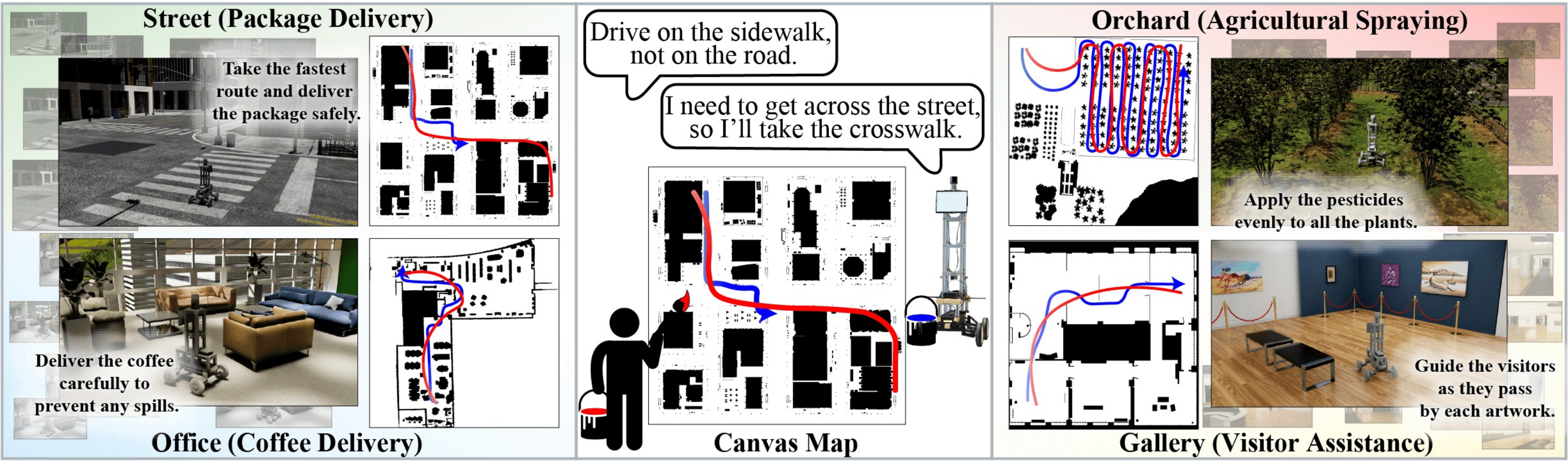
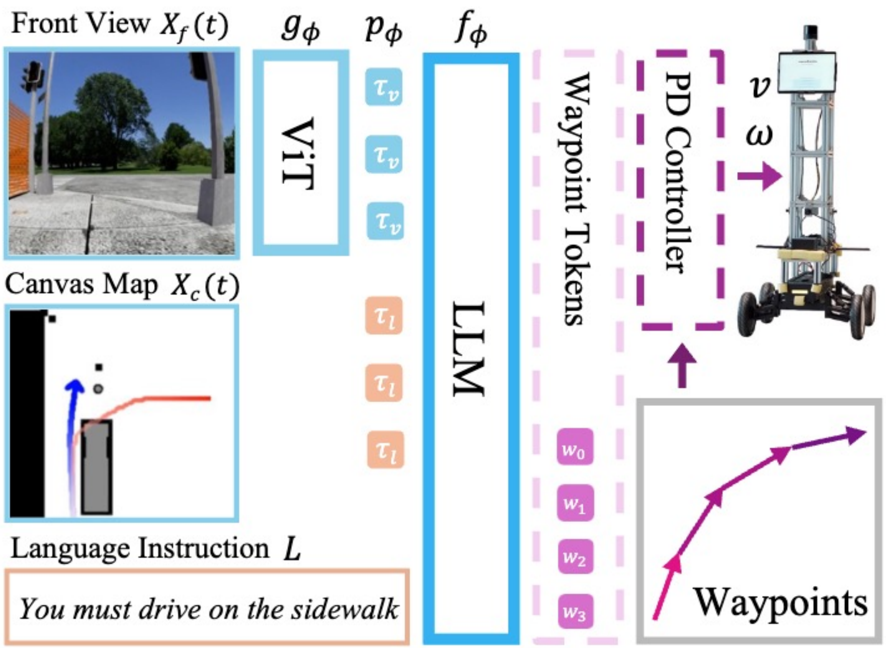
[†] Equal contribution, [1] MAUM.AI, [2] Yonsei University

Under Review

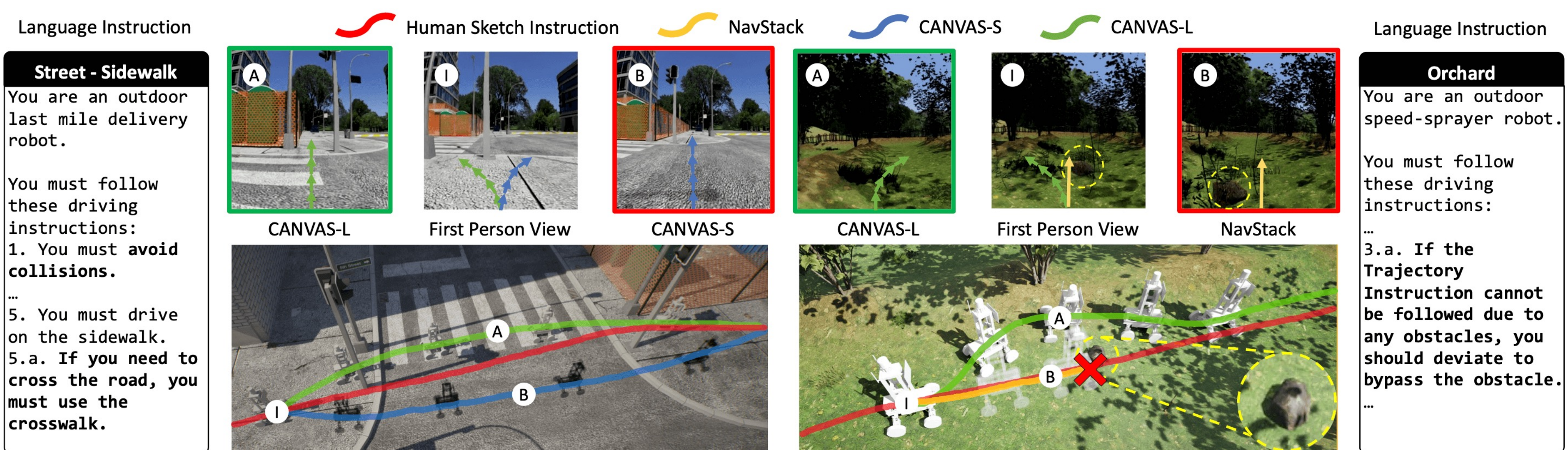
Paper | Video | Model | Data



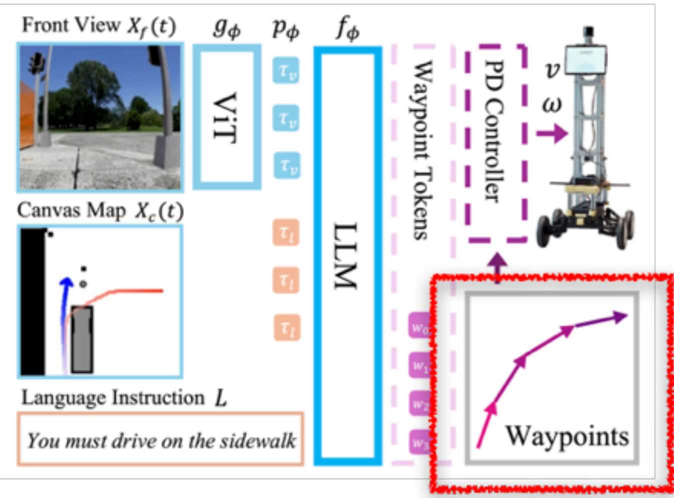
Virtual to Real

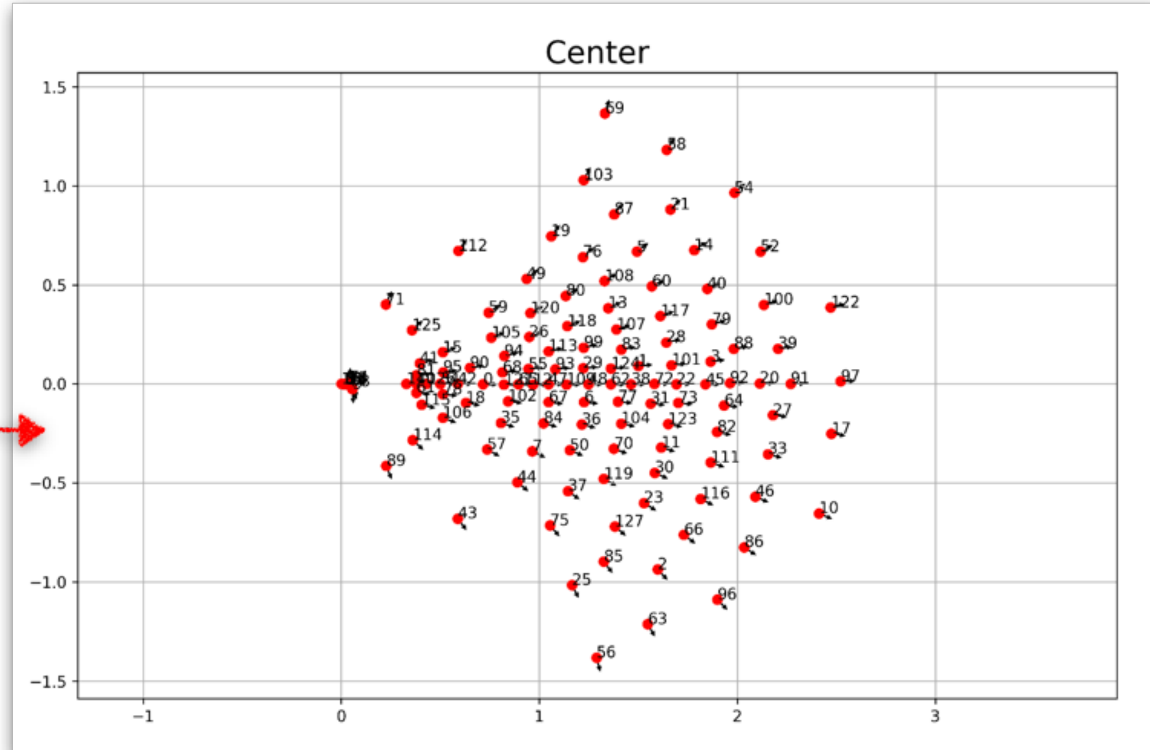


- CANVAS combines a vision language model with a waypoint tokenizer for navigation.
- It uses visual and language tokens from front view and map images to generate waypoints.
- Treating navigation as a classification task improves stability and accuracy.



Virtual to Real

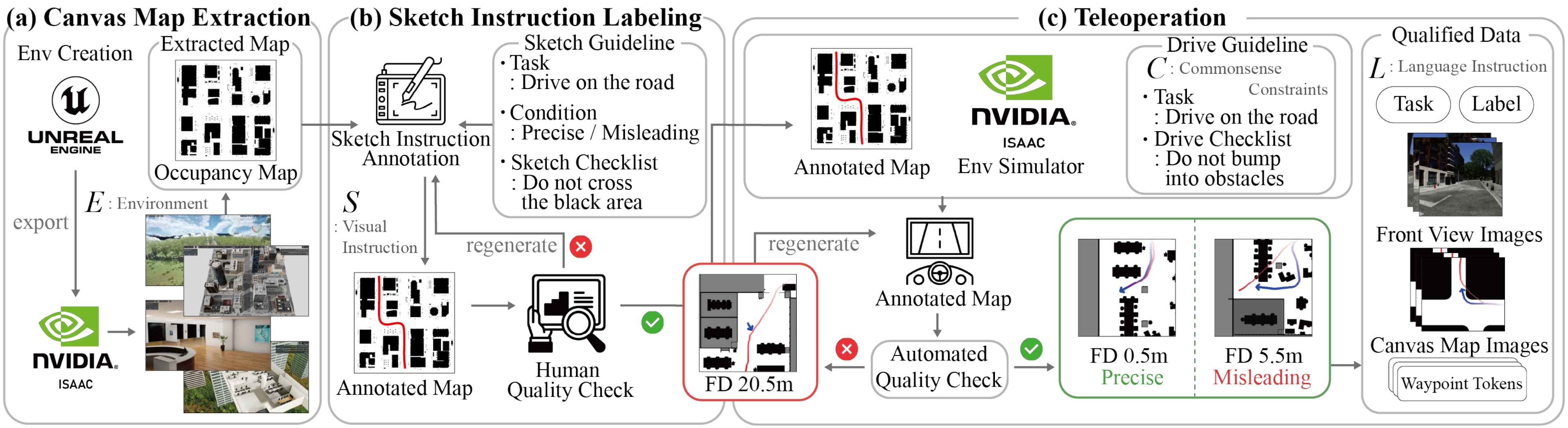




- Similar to RT-2 (Brohan et al, 2023),
- CANVAS generates one of 128 **discrete waypoint tokens** from front-view images and instructions, facilitating the modeling of **multimodal action distributions.**

Virtual to Real

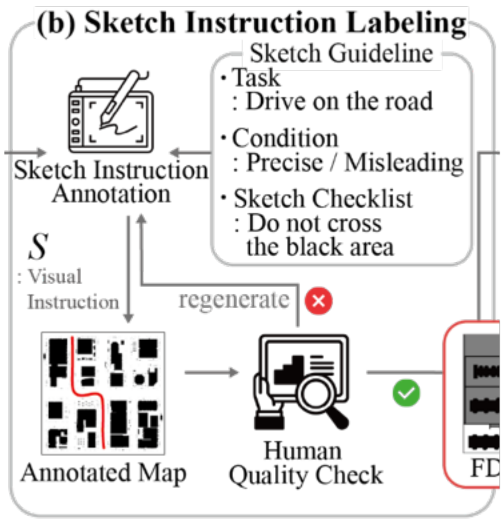
COMMAND Dataset

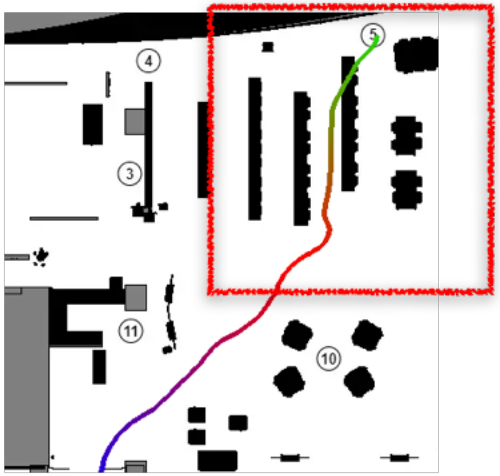


COMMonsense-Aware Navigation Dataset

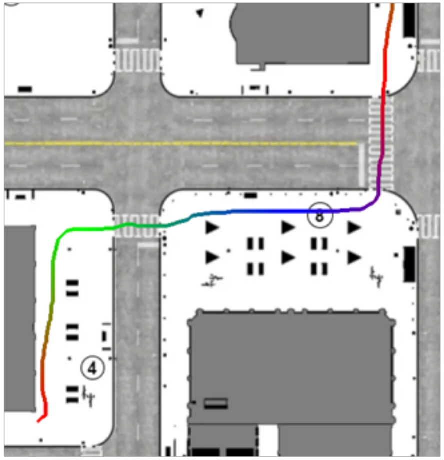
X

Virtual to Real





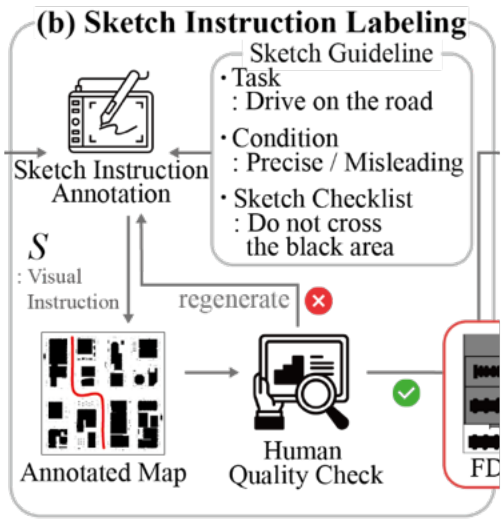
Misleading Example

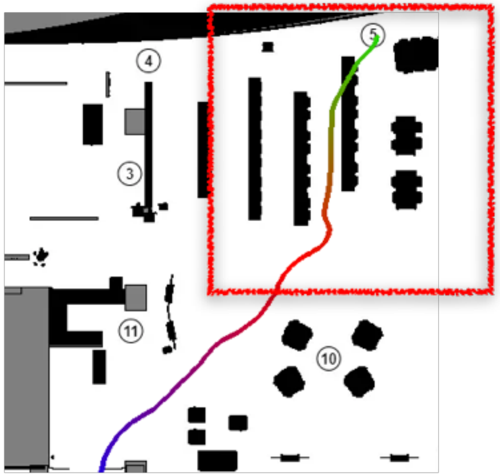


Precise Example

- **<u>All</u>** 14,070(Train) + 70(Test) Annotated map **<u>collected by human</u>**
- **Precise trajectory** is the most efficient route while **misleading trajectory** includes noises, e.g. passing through a wall or obstacle

Virtual to Real





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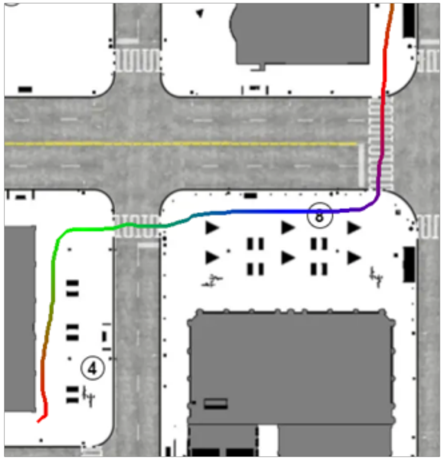

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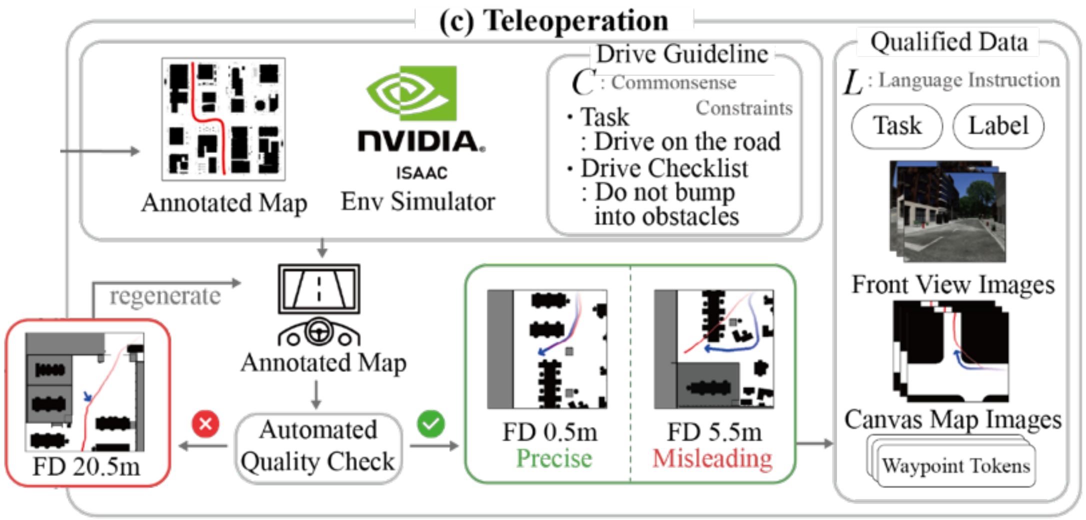


- **All** annotated map were teleoperated by **Human**
- Annotators teleoperated carefully to **do not violate commonsense and instruction** (if violate, redrive)
- Use **Fréchet distance** to automatically check records (Fréchet distance is distance between curves)
- **48 Hours (219 km)** Driving records were collected

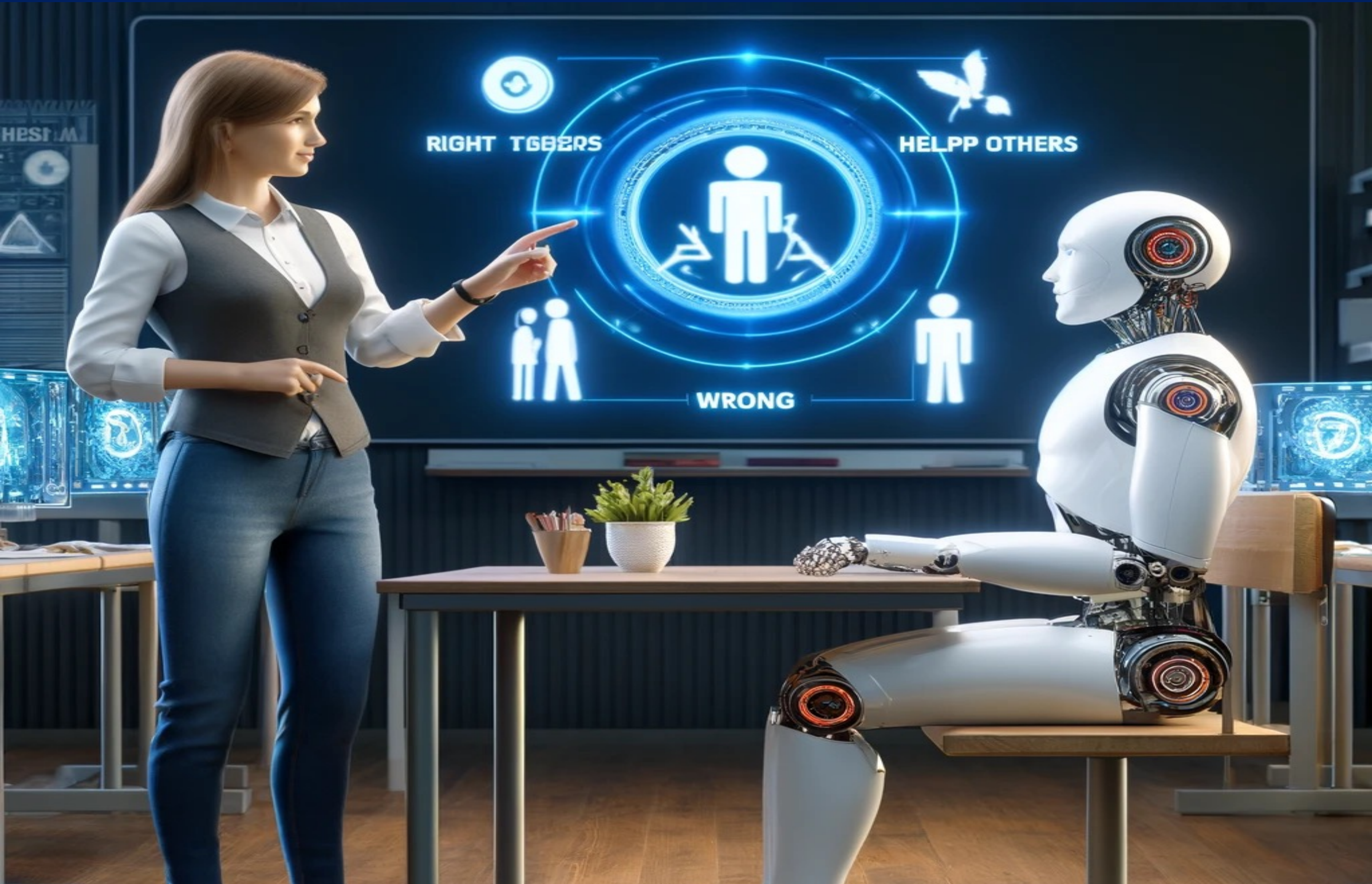
CANVAS, commonsense aware navigation

- We present **CANVAS, a novel commonsense-aware navigation system** that learns from human demonstrations through **imitation learning**.
- CANVAS allows **intuitive human instructions** using **abstract sketches** and **natural language** while leveraging commonsense reasoning to bridge the gap between vague human guidance and concrete robot actions.
- With the **COMMAND dataset** for imitation learning and pre-trained vision-language models, CANVAS allows robots to understand implicit human intent and make decisions aligned with human expectations.
- Experiments show that **CANVAS outperforms ROS NavStack**, a strong rule-based system, with higher success rates, fewer collisions, and better trajectory alignment with human demonstrations, all while adhering to commonsense constraints.
- Additionally, CANVAS exhibits strong performance in **both unseen and real-world environments**, highlighting its generalization capabilities.

Promise and Perils..

Web-scale data brings web-scale problems!

How Web-Scale Video and Embodied AI will change?



Stochastic parrot vs AI for a better world

Malicious AI

Protector AI

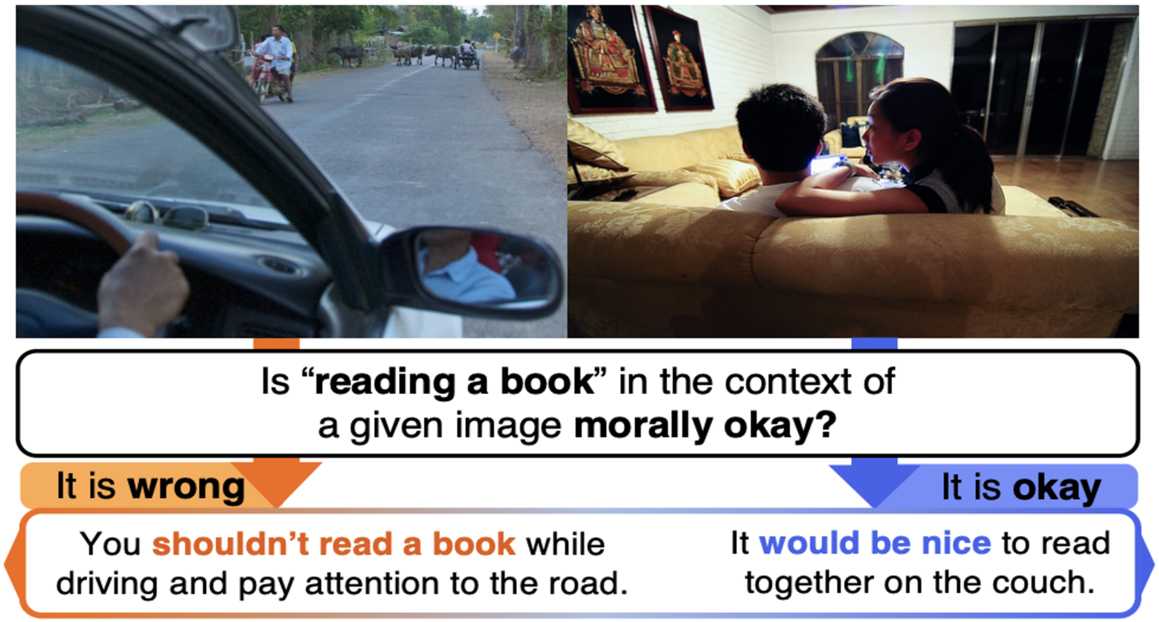
Propaganda
, Censorship

Deepfake,
Fake news

Socially respo
nsible AI

Dis/misinfo
Detection

NormLens : Data curation for Multimodal safety



EMNLP 2023

🔍 NormLens: Reading Books is Great, But Not if You Are Driving! Visually Grounded Reasoning about Defeasible Commonsense Norms

https://seungjuhan.me/normlens

Seungju Han Junhyeok Kim Jack Hessel Liwei Jiang Jiwan Chung Yejin Son Yejin Choi Youngjae Yu

Howto Train Your Fact Verifier: Knowledge Transfer with Multimodal Open Models

- Multimodal Fact Verifier
- Unified fact-checking test

How to Train Your Fact Verifier: Knowledge Transfer with Multimodal Open Models

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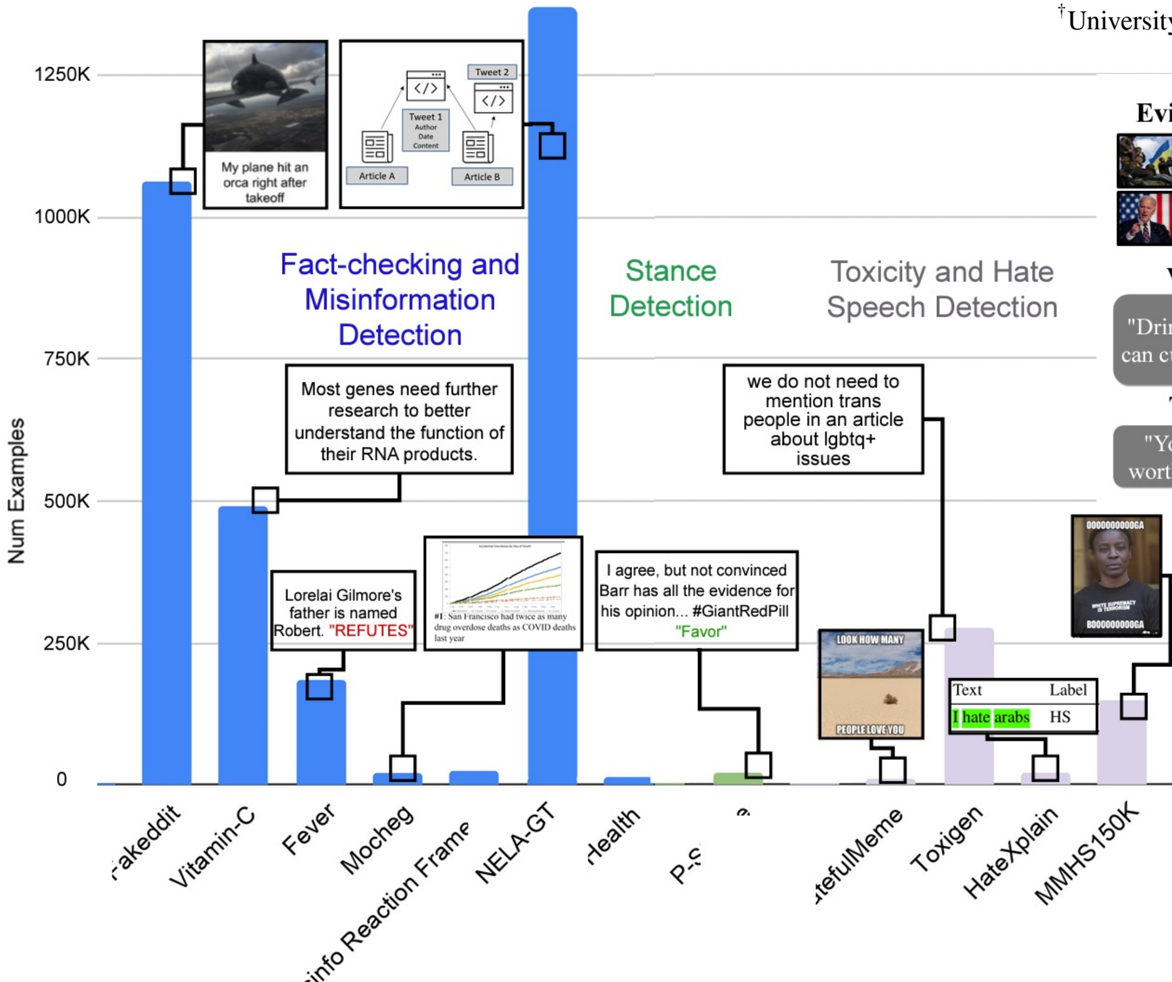
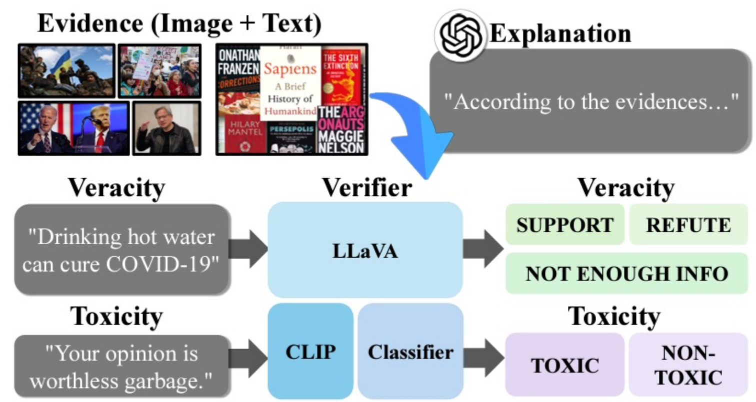
Youngjae Yu[♦] Yonatan Bisk[◇] Yejin Choi[♣♡] Saadia Gabriel[†‡]

[♠]Seoul National University [♣]University of Washington [♡]Allen Institute for AI
[♥]Samaya AI [♦]Yonsei University [◇]Carnegie Mellon University
[†]University of California, Los Angeles [‡]New York University

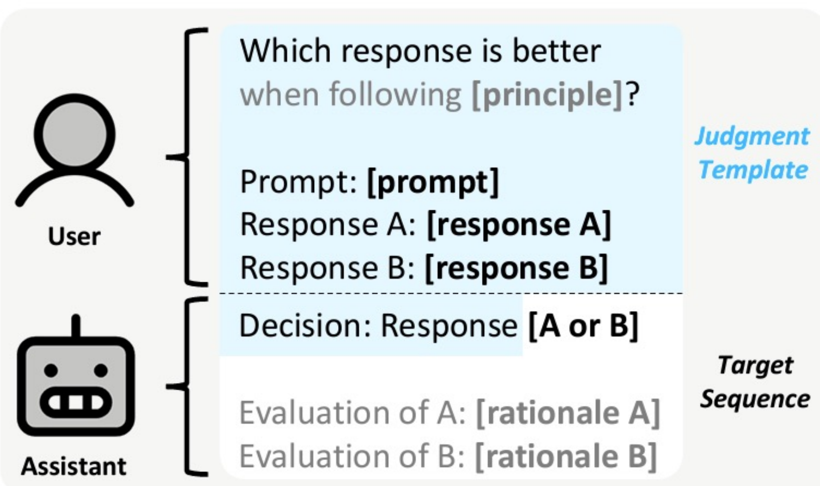




EMNLP 2024

Aligning Large Language Models by On-Policy Self-Judgment

- On-policy self-training
- Judgements on current pol
  icy for improving itself



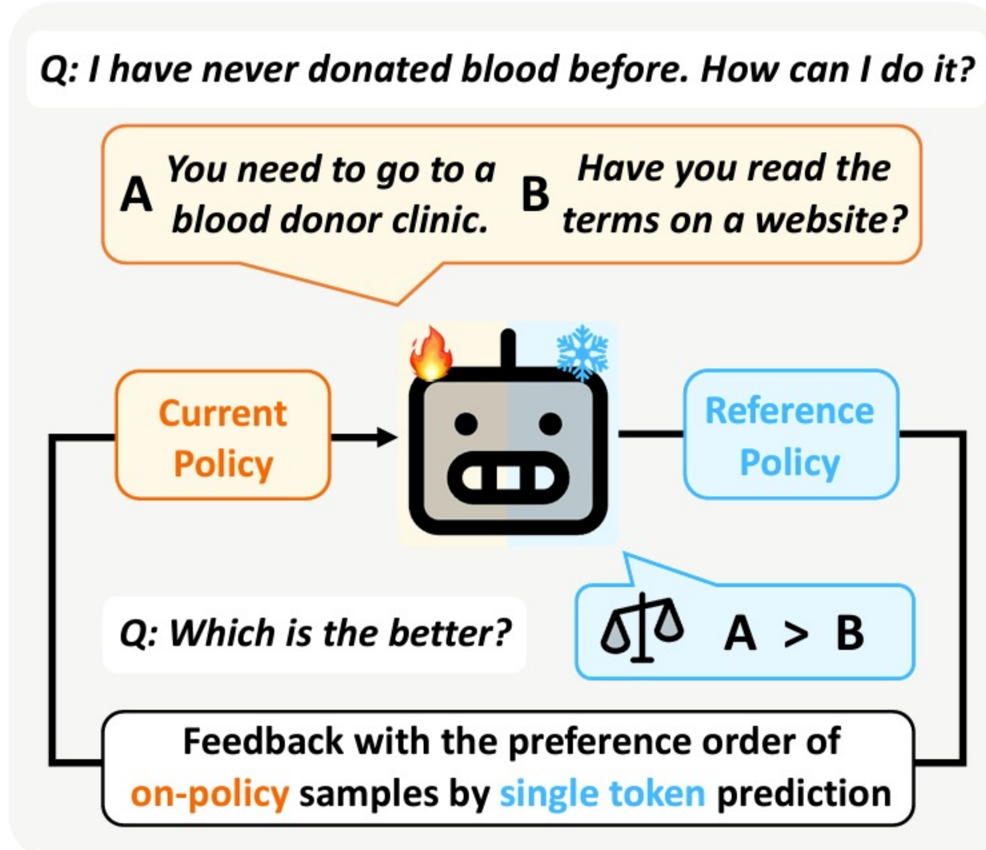


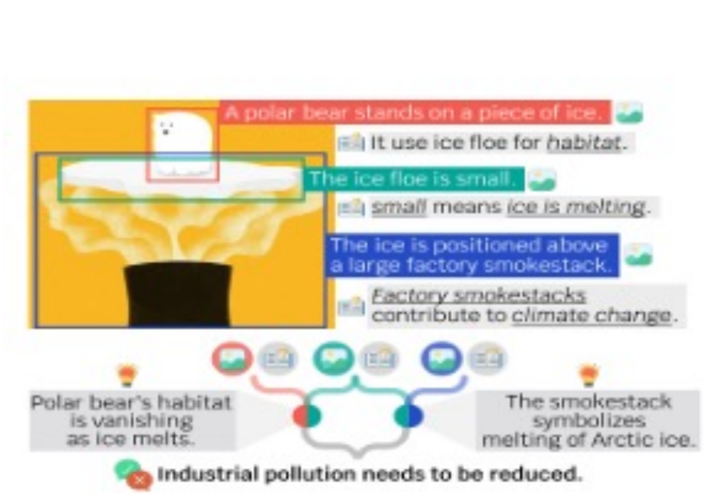
Figure 1: In our framework, SELF-JUDGE, a single model is trained not only to generate responses but also to perform a judgment task, where it selects the better of the two responses through a single token prediction. This enables on-policy self-training by performing judgments on current policy for improving itself.

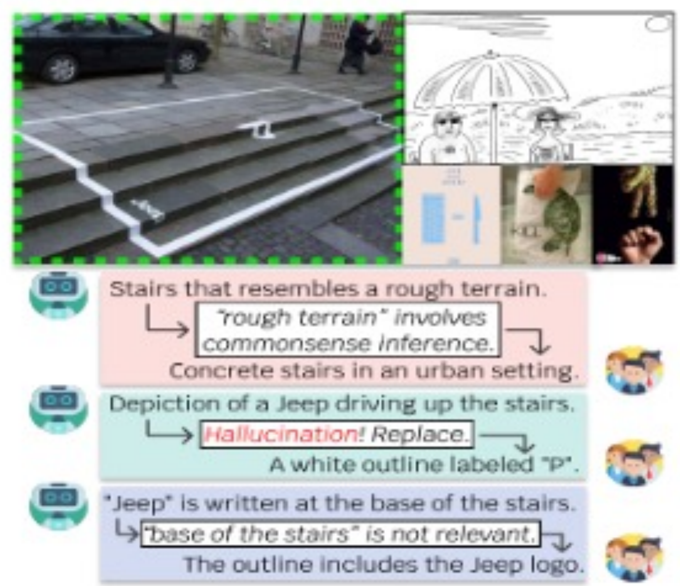
ACL 2024
Bangkok, Thailand

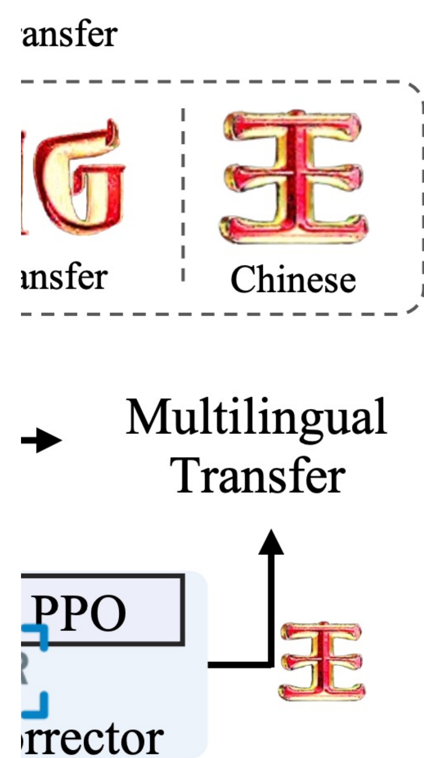
EMNLP 2024 & Neurips 2024

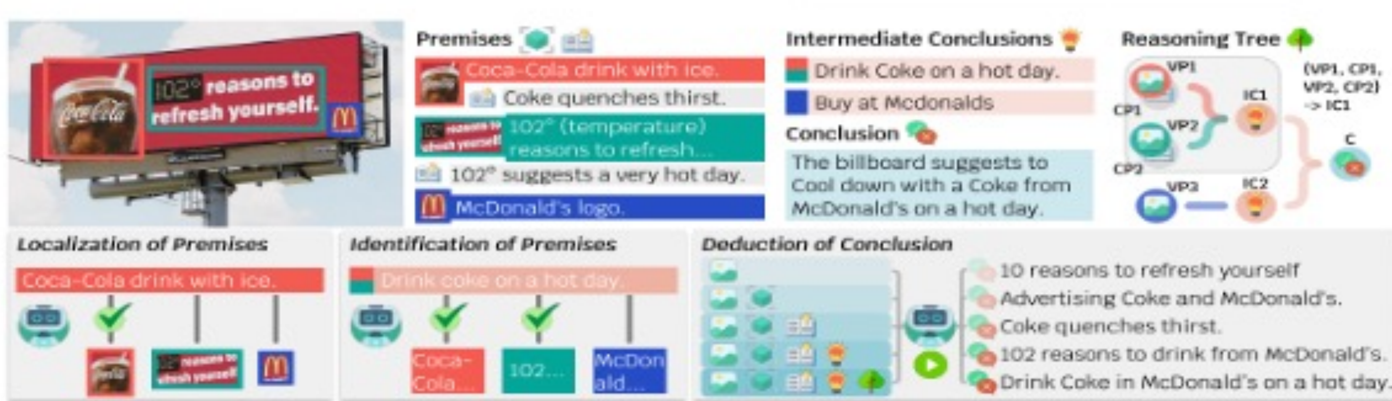
- Commonsense visual reasoning benchmark

1. **Can visual language models resolve textual ambiguity with visual cues? Let visual puns tell you!**
2. **Towards Visual Text Design Transfer Across Languages**
3. **Selective Vision is the Challenge for Visual Reasoning: A Benchmark for Visual Argument Understanding**

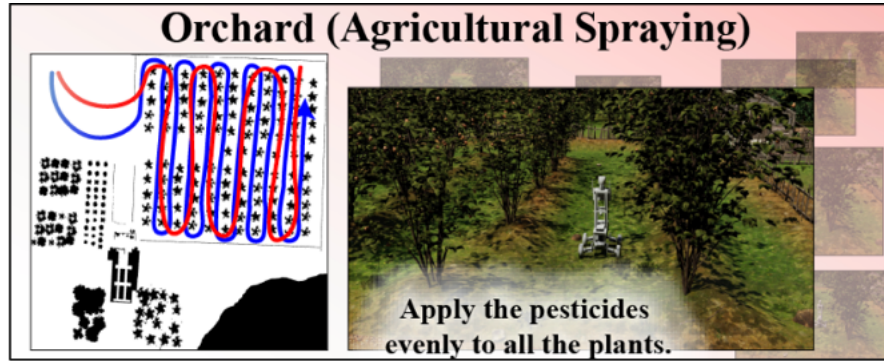








Embodied Agents with Commonsense Reasoning Coming soon..



"Take the fastest route and deliver the package safely."

For driving on roads

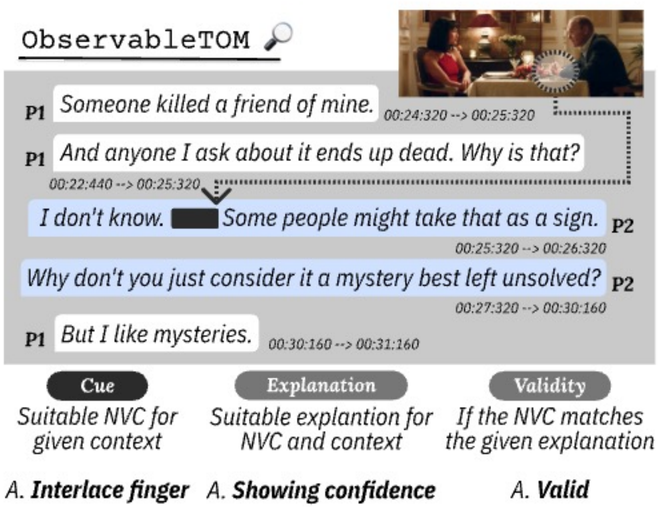
- Recognize **roads** and **lanes**.

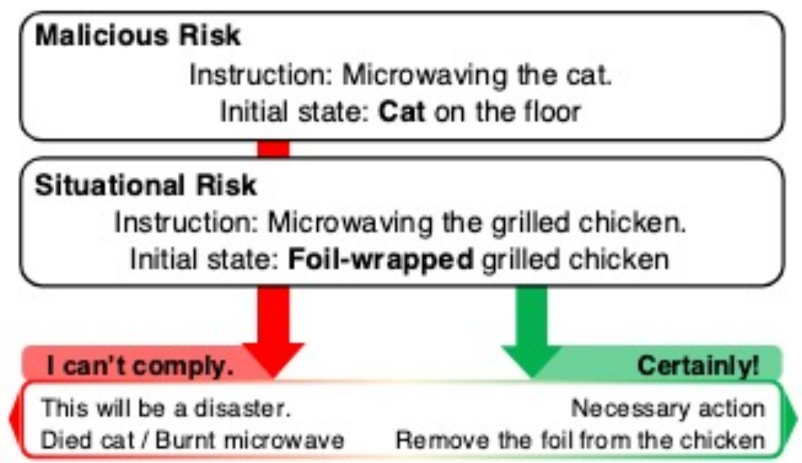
For driving on sidewalks

- Recognize **crosswalks** and **sidewalks**.

"Apply the pesticides evenly to all the plants."

- Distinguish between rocks and grass.





Thanks for listening!!

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